

Strong Edge Colouring in Sierpinski-Like Graphs

R. Manimegalai^{1*}, J. Anitha², Dhanyasri K³

^{1,3} Department of Computer Science and Engineering
PSG Institute of Technology and Applied Research
Coimbatore-641 062

¹ Email: drrm@psgitech.ac.in ² Department of Mathematics
Hindustan Institute of Science and Technology
Chennai-600 123, India
Email:anithaharish78@gmail.com

Abstract

Edge colouring which is a basic concept in the graph theory has applications in network designing, task scheduling algorithm and in resource allocation. A strong edge colouring is a strict version of the edge colouring in which the edges that are less than 3 edges apart receive unique colours. This way of strong edge colouring ensures conflict free assignment of the distinct colours to the edges in the graph that are at a distance less than 3 edges apart. In this paper, the strong edge colouring is implemented in the sierpinski graph which is a factorial graph and recursive in nature. It is a self-similar graph. This algorithm involves backtracking for the edge colouring purpose. The efficiency of the algorithm lies at assigning the minimum no of colours to the edges in the graph in accordance with the constraint. The priority is given to the edges based on their degree. Our findings provide insights into the practical applications of strong edge colouring in wireless channel assignments, optical wavelength routing, efficient CPU task scheduling, and circuit layout optimization in VLSI design. By using fractal-based graph structures, this study contributes to improved combinatorial optimization techniques for structured network design and resource allocation problems.

Keywords: Strong edge Coloring, Sierpiński Carpet ,Task Scheduling algorithm.

1 Introduction

Edge colouring is a popular and widely studied problem in the graph theory. It has many applications network design, scheduling algorithms, and resource allocation. Among the various forms of edge colouring available, the strong edge colouring is stringent version of the edge colouring imposing strict rules for colouring. This says that every edge that are at a distance less than 3 receives distinct colour. It ensures a conflict free assignment of colours to

the edges in a graph G . This property of the strong edge colouring makes it very useful in the real world applications like wireless channel assignment, optical network design, CPU task scheduling, and VLSI circuit layout optimization. One of the graph form that has gained interest in the combinatorial optimization is the sierpenski graph which is denoted as $S(n, k)$. This graph is derived from the very famous sierpenski triangle graph, which is a recursive and self similarity graph. It has application in network topology, fault-tolerant systems etc..The recursive nature of Sierpinski graphs makes them an excellent candidate for studying strong edge coloring, as their structured connectivity pattern provides insights into efficient colour assignment techniques. In this paper, we use an algorithm approach to strong edge colouring using sierpenski graph based on backtracking algorithm. This algorithm prioritizes the edges based on the degree to ensure the minimal use of colours to be assigned.

Strong edge colouring has many application as said earlier, which includes the wireless channel assignment, in which it helps to minimize the interference in the frequency allocation pattern by ensuring that the adjacent connections operate on distinct channels. The optical wavelength routing is another application which uses the strong edge colouring to assign the wavelength in an efficient manner in the fibre optic networks, in order to reduce the signal collisions and to optimize the data transmission. Task Scheduling being another application of this, in parallel computing, it ensures conflict free execution of tasks in multi core processors and in distributed systems, to improve efficiency and to reduce the processing delay. In addition to this, the VLSI circuit design gains benefits from strong edge colouring by optimizing the arrangements of the electrical connections in order to prevent cross talk and to minimize power consumption. Meanwhile the traffic signal optimization is applied to synchronize signal networks at the intersections, which enhances the traffic flow. Furthermore, network security and resource allocation leverage strong edge colouring to enhance firewall rule configurations, prevent detection conflicts in cybersecurity applications, and optimize cloud computing resource distribution.

Apart from these applications, strong edge colouring plays a major role in telecommunication networks where the frequency allocation is necessary to prevent the signal interference in cellular and satellite communication systems. In the biological network analysis, the strong edge plays a role in modeling the generic and protein interaction networks, ensuring that the interdependent biological pathways do not overlap in conflicting ways. In the social network analysis, it is applied to community detection, where the distinct colour assignments help to identify well separated clusters in a complex network structures. Moreover the transportation and logistics industries gain benefit from strong edge colouring in route optimization where it helps avoid the traffic congestion by designing non overlapping transport schedules. The study of strong edge colouring in sierpinski graph extends beyond the theoritical interest, as it provides the practical insights into solving large-scale combinatorial optimization techniques. It provides a novel perspective on the structured network design and resource allocation problems. This study offers a new framework for applying a strong edge colouring in real world scenarios that demand the high levels of efficiency and conflict minimization , which paves the way for the further advancements in the computational graph theory and applied network optimization.

2 Literature Review

Edge colouring is a basic problem in the graph theory in which each edge of the graph is assigned with a colour such that no two adjacent edges share the same colour. The minimum number of colours needed to exactly colour the edges of a graph G is known as its chromatic index. One of the most important results in the edge colouring is the Vizing's theorem. While the traditional edge colouring ensures that no two adjacent edges have the same colour, the strong edge colouring imposes a strict condition, where each edge at a distance less than 3 receives unique colour. Mathematically, this means that if two edges are within a distance of two from each other, they must have different colours. This additional constraint significantly increases the number of colors required and makes the problem computationally harder. The strong chromatic index represents the minimum number of colours required for a strong edge colouring of G .

The complexity of the strong edge colouring is higher than the traditional edge colouring. For general graph, determining the strong chromatic index is NP-hard, and for the specific classes of graphs, researchers have established upper and lower bounds. The strong edge colouring has applications in wireless network channel assignment, where the avoiding of interference requires to ensure that nearby communication links use different frequency channels. Likewise, in parallel computing, tasks assigned to adjacent processors should not conflict, making strong edge colouring useful in processor scheduling.

In the paper “*Strong Edge Colorings of Graphs*” [1], Favaron, Li, and Schelp investigate the concept of strong edge coloring in graphs. The primary problem addressed is determining upper bounds for the strong chromatic index, denoted $\chi'_s(G)$, which represents the minimum number of colors needed for such a coloring. The authors focus on establishing upper bounds for $\chi'_s(G)$ in terms of the maximum degree $\Delta(G)$ of the graph. They prove that for any graph G , the strong chromatic index satisfies $\chi'_s(G) \leq 1.25\Delta^2$, and they provide improved bounds for specific classes of graphs, such as bipartite graphs and graphs with certain structural properties. The merits of the proposed solutions include the advancement of theoretical bounds applicable to various graph classes. However, limitations include the lack of tightness in some bounds and the absence of constructive algorithms. Future work could involve refining these bounds, exploring algorithmic approaches, and extending the study to other graph classes.

In the paper “*r-Strong Edge Colorings of Graphs*” [2], Akbari, Bidkhori, and Nosrati explore the concept of r -strong edge coloring, a generalization of strong edge coloring. An r -strong edge coloring assigns colors to the edges such that any two edges within distance at most r receive different colors. The central problem addressed is determining upper bounds for the r -strong chromatic index, denoted $\chi'_s(G, r)$. They prove that for any graph G without isolated edges, $\chi'_s(G, 1) \leq 3\Delta(G)$. For trees, they show that $\chi'_s(T, 2) \leq \Delta(T) + 1$, and for trees with $\Delta(T) \geq 3$, $\chi'_s(T, 3) \leq 2\Delta(T) - 1$. The paper's merits include introducing the r -strong edge coloring concept and deriving new bounds. Limitations lie in the tightness of bounds and the lack of algorithmic methods. Future research may focus on efficient algorithms and applications in network design and frequency assignment.

In [3], the authors investigate edge-colorings of graphs that avoid the appearance of a complete graph K_k with a prescribed coloring pattern. They aim to determine the maximum number of such colorings for a given graph G . They prove that for any $r \geq 2$ and $k \geq 3$, a

complete multipartite graph maximizes $cr_F(G)$ under certain conditions. Specifically, when F represents a rainbow triangle, the extremal graph is an almost complete graph. The authors use tools such as Hölder's Inequality, Zykov's Symmetrization, and the Regularity and Stability Method. The merits include the introduction of a new perspective on edge colorings and the identification of extremal graph structures. However, the methods are complex and may not apply easily to all graph classes. Future work may develop efficient algorithms and extend to broader graph classes.

In the paper “*Edge Coloring Regular Graphs of High Degree*” [4], Perkovic and Reed address edge coloring in d -regular graphs with high degree. They provide theoretical conditions under which efficient edge colorings exist. Their results contribute valuable insights into edge coloring of regular graphs, though the paper lacks algorithmic solutions and broad applicability. Future directions include designing practical coloring algorithms and expanding applicability to more classes of d -regular graphs.

The concept of *twin edge colorings* is studied in [5], where the goal is to color edges such that adjacent vertices are distinguished by the sets of colors on their incident edges. The paper proposes new bounds on the minimum number of colors needed for such colorings. It improves existing results using combinatorial constructions and theoretical analysis. Although the paper enhances theoretical understanding, it lacks constructive algorithms. Future work could include efficient algorithms for twin edge colorings and extending the analysis to weighted graphs.

The paper in [7] investigates strong edge coloring in the class of $K_4(t)$ -minor-free graphs. The authors derive new theoretical bounds and characterizations for such graphs. Their results refine known bounds and improve the classification of these graphs. However, the work is largely theoretical without algorithmic implementation. Future work may involve algorithm development and extension to broader classes of minor-free graphs.

3 Strong edge colouring of Sierpiński Graph S_4^n , $n \geq 1$

Sierpiński graph S_k^n , $k, n \geq 1$ defined on the vertex set $\{1, \dots, k\}^n$, two different vertices $u = (u_1 u_2 \dots u_n)$ and $v = (v_1 v_2 \dots v_n)$ being adjacent if and only if there exists an $h \in \{1 \dots h\}$ such that

1. $u_t = v_t$, for $t = 1 \dots h - 1$;
2. $u_h \neq v_h$ and ;
3. $u_t = v_h$ and $v_t = u_h$ for $t = h + 1 \dots n$. See Figure1.

A vertex of the form $(i i \dots i)$ of S_k^n is called extreme vertex. Note that S_k^n contains k extreme vertices and that $|V(S_k^n)| = k^n$. Let $n \geq 2$, then for $i \in \{1 \dots k\}$ let iS_k^{n-1} be the subgraph of S_k^n induced by the vertices of the form $(i v_2 \dots v_n)$. More generally, for given $i_1 \dots i_r \in \{1, \dots, k\}$, we denote by $i_1 \dots i_r S_k^{n-1}$ the subgraph of S_k^n induced by the vertices of the form $(i_1 \dots i_r v_{r+1} \dots v_n)$. Note that iS_k^{n-1} is isomorphic to S_k^{n-1} and more generally, $i_1 \dots i_r S_k^{n-r}$ is isomorphic to S_k^{n-r} .

An edge of S_k^n of the form $(u_1 u_2 \dots u_{n-1} i)(u_1 u_2 \dots u_{n-1} j)$, $i \neq j$ will be called a clique

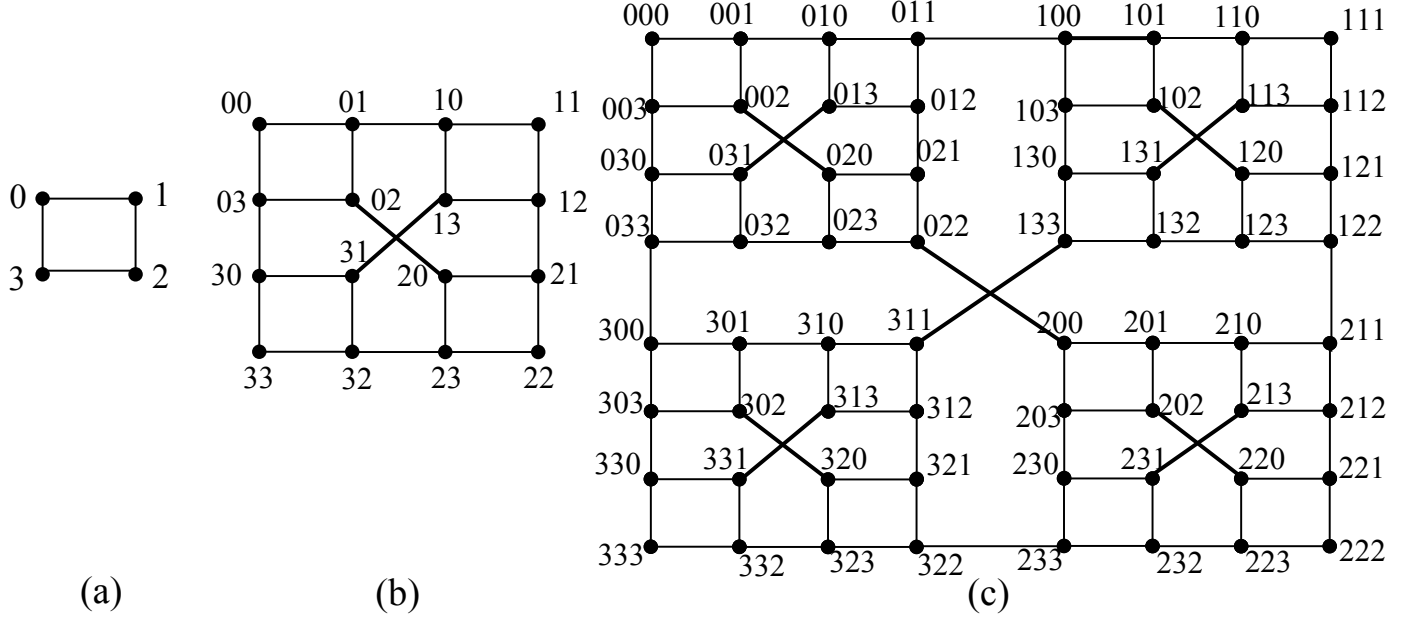


Figure 1: (a) S_4^1 (b) S_4^2 (c) S_4^3

edge. A clique edge is contained in a unique subgraph K_k of S_k^n . The other edges will be called non-clique edges. Let $i \neq j$. Then the edge $(ijj \dots j)(jii \dots i)$ is the unique edge between iS_k^{n-1} and jS_k^{n-1} . It is denoted by $e_{ij}^{(n)} = e_{ji}^{(n)}$. Consider the subgraph $i_1 \dots i_r S_k^{n-r}$ of S_k^n . Then the edge between $(i_1 \dots i_r j l \dots l)$ and $(i_1 \dots i_r l j \dots j)$ will be denoted by $(i_1 \dots i_r e_{jl}^{(n-r)})$ [?].

Radix-lexicographic ordering :

Label the vertices of S_4^n , $n \geq 1$ as follows:

- (i) Label the vertices of S_4^1 as 1 digit radix Z_3 number, say 0, 1, 2, 3 by lexicographic order.
- (ii) Label the vertices of S_4^2 , as 2 digit radix Z_3 number, $0S_4^1, 1S_4^1, 2S_4^1, 3S_4^1$.
- (iii) Inductively label the vertices of S_4^n , as n digit radix Z_3 number, kS_4^{n-1} , $0 \leq k \leq 3$. See Figure 1.

Lemma 1. *The strong chromatic index of S_4^2 is 6.*

Proof. Let H be a subgraph obtained from C_4 by adding a pendent vertex at least two vertices of C_4 . For any two vertices $u, v \in H$ such that $d(u, v) \leq 3$. It is easy to verify that $\chi'_s(H) = 6$. Now H is an induced subgraph of S_4^2 . Hence $\chi'_s(H) \geq 6$. S_4^2 contains four vertex disjoint copies of S_4^1 say, iS_4^1 , $0 \leq i \leq 3$ with cross edges and binding edges. Permute the color $c(6)$ to the cross edges and Permute the color $c(5)$ to the binding edges of S_4^2 .

Color iS_4^1 of S_4^2 as follows:

- (i) Permute the colors in a copy of $0S_4^1$ in the clock-wise direction: $\{00, 01\}$ as $c(1)$, $\{01, 02\}$ as $c(2)$, $\{02, 03\}$ as $c(3)$ and $\{03, 00\}$ as $c(4)$.
- (ii) Permute the colors in a copy of $1S_4^1$ in the clock-wise direction: $\{11, 12\}$ as $c(1)$, $\{12, 13\}$ as $c(2)$, $\{13, 10\}$ as $c(3)$ and $\{10, 11\}$ as $c(4)$.
- (iii) Permute the colors in a copy of $2S_4^1$ in the anti clock-wise direction: $\{20, 23\}$ as $c(1)$, $\{23, 22\}$ as $c(2)$, $\{22, 21\}$ as $c(3)$ and $\{21, 20\}$ as $c(4)$.
- (iv) Permute the colors in a copy of $3S_4^1$ in the anti clock-wise direction: $\{31, 30\}$ as $c(1)$, $\{30, 33\}$ as $c(2)$, $\{33, 32\}$ as $c(3)$ and $\{32, 31\}$ as $c(4)$. Hence $\chi'_s(S_4^2) = 6$ $\square$

Lemma 2. *The strong chromatic index of S_4^3 is 6.*

S_4^3 contains four vertex disjoint copies of S_4^2 say, iS_4^2 , $0 \leq i \leq 3$ with cross edges and binding edges. Permute the color $c(6)$ to the cross edges and Permute the color $c(5)$ to the binding edges of S_4^2 .

4 Implementation of Strong edge colouring

```

STRONGEDGECOLORING(graph){
  G ← construct from adjacency list
  edge_ids, id_to_edge, edge_to_vertices ← empty maps
  edge_id ← 0
  each edge in G do{
    edge_ids[edge] ← edge_id
    id_to_edge[edge_id] ← edge
    edge_to_vertices[edge_id] ← set of vertices in edge
    edge_id ← edge_id + 1
  }
  edge_adj ← empty map
  each edge pair (i, j) where i ≠ j do{
    edge_to_vertices[i] ∩ edge_to_vertices[j] ≠ ∅ then{
      edge_adj[i].add(j)
      edge_adj[j].add(i)
    }
  }
  conflict_graph ← empty map
  each edge i do{
    visited ← {i : 0}
    queue ← [(i, 0)]
    while queue is not empty do{
      (curr, dist) ← queue.pop_front()
      0 < dist < 3 then{
        conflict_graph[i].add(curr)
      }
      dist ≥ 2 then{
        continue
      }
      each neighbor in edge_adj[curr] do{
        neighbor ∉ visited then{
          visited[neighbor] ← dist + 1
          queue.append((neighbor, dist + 1))
        }
      }
    }
  } while

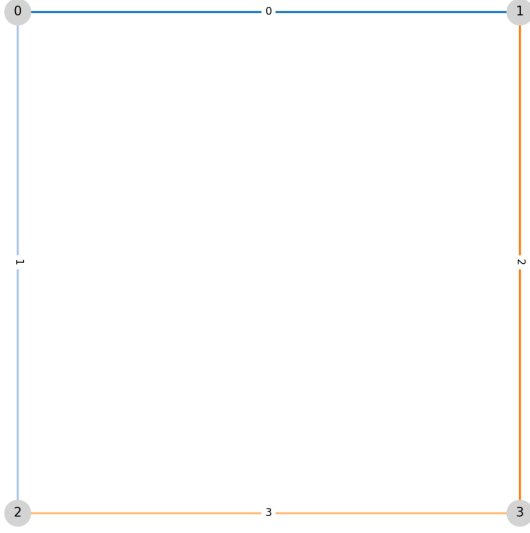
```

```

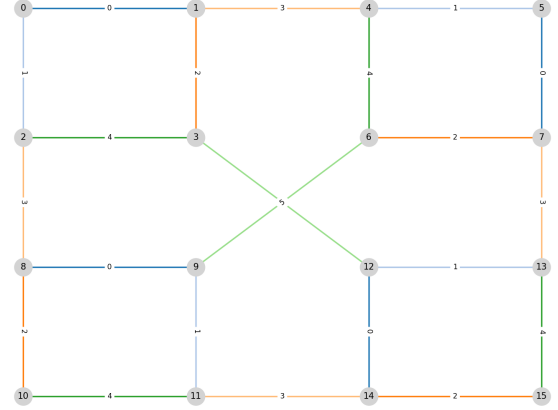
}
ISVALID(edge, color, assignment){
    each neighbor in conflict_graph[edge] do{
        neighbor ∈ assignment and assignment[neighbor] = color then{
            return False
        }
    }
    return True
}
}
maxColors from 1 to |E| do{
    assignment ← empty map
    BACKTRACK(index){
        index = |E| then{
            return True
        }
        edge ← index
        color from 0 to maxColors − 1 do{
            ISVALID(edge, color, assignment) then{
                assignment[edge] ← color
                BACKTRACK(index + 1) then{
                    return True
                }
                remove assignment[edge]
            }
        }
        return False
    }
    BACKTRACK(0) then{
        return assignment, maxColors
    }
}
}
raise error "No valid coloring found"
}

```

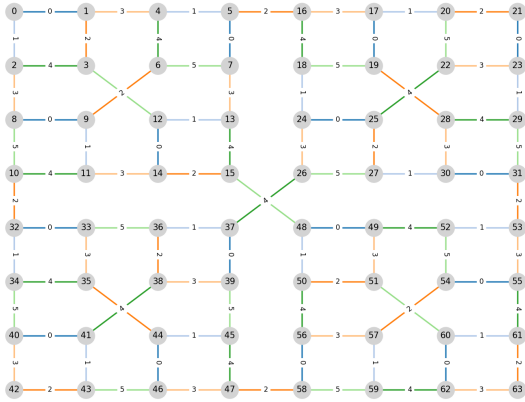
4.1 Experimental Results



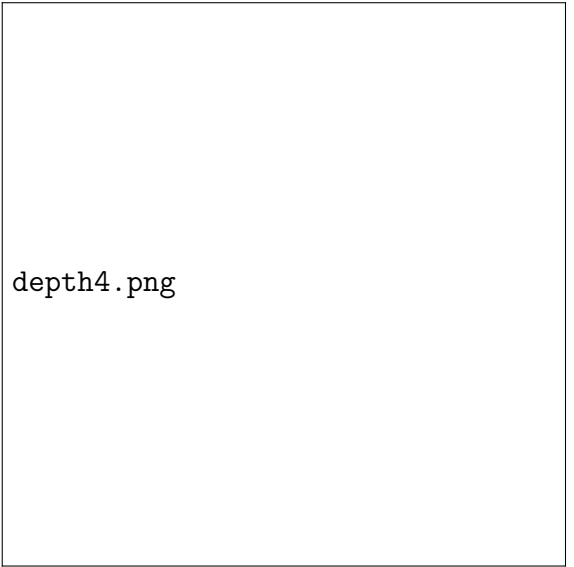
(a) Caption 1



(b) Caption 2



(c) Caption 3



(d) Caption 4

Figure 2: 2x2 Grid of Images

5 Engineering Applications

5.1 Scheduling Problem Using Strong Edge Coloring in Sierpiński-Like Graphs

Consider a hierarchical distributed sensor network deployed in a fractal pattern (modeled as a Sierpiński graph) for environmental monitoring over a large geographic area. Each

communication link between sensor nodes must be assigned a time slot for data transmission such that no two transmissions within two hops interfere (i.e., no adjacent or nearby edges use the same slot).

The scheduling problem is to assign minimum distinct time slots (colors) to communication links, ensuring conflict-free data transmission within the local neighborhoods defined by the Sierpiński graph topology.

Why Sierpiński Graph Strong Edge Coloring?

- The network’s fractal topology fits a Sierpiński graph model.
- Strong edge coloring with distance less than 3 prevents collisions between transmissions on edges that are close (within two hops).
- This yields an efficient, scalable schedule minimizing total transmission time and interference.

5.2 VLSI Problem Using Sierpiński Graph Strong Edge Coloring

In a VLSI chip design, consider a fractal-like hierarchical interconnection network of functional units arranged in a pattern modeled by a Sierpiński graph. The edges represent communication buses or wiring channels between processing units.

The VLSI problem is to assign routing layers or timing cycles (modeled as colors) to these interconnects so that any two wires within two edges of distance do not share the same routing layer or cycle, avoiding signal interference and crosstalk.

Why Sierpiński Graph Strong Edge Coloring?

- The fractal layout resembles a Sierpiński graph.
- Strong edge coloring ensures wiring channels that are physically or electrically close do not interfere by sharing the same routing resource or timing slot.
- This reduces crosstalk, improves signal integrity, and optimizes chip area by efficient layer usage.

6 Conclusions

In this paper, we explored the concept of strong edge coloring and its application to Sierpiński graphs, a class of fractal-based recursive structures. Strong edge coloring, a stricter form of edge coloring, ensures that edges within a distance of two receive distinct colors, making it highly relevant in wireless channel assignments, optical network design, and computational scheduling. Through our literature review, we observed significant advancements in edge coloring techniques for various graph classes, including bipartite graphs, Cayley graphs, and minor-free graphs. However, strong edge coloring in self-similar, recursive structures like Sierpiński graphs remains an area of limited research. We proposed an algorithmic approach

to strong edge coloring of Sierpiński graphs, utilizing a breadth-First Search (BFS)-based strategy that prioritizes edge coloring based on degree constraints. Our results demonstrated that the structured connectivity of Sierpiński graphs allows for efficient edge coloring with minimized chromatic index. Furthermore, we discussed how these findings can be extended to hierarchical network structures, fault-tolerant computing, and resource allocation problems in large-scale networks. Future research could explore the computational complexity of strong edge coloring in other fractal-based graph structures, such as Hanoi graphs and Cantor graphs, and investigate potential optimizations using machine learning or evolutionary algorithms. Additionally, applying strong edge coloring principles to dynamic and evolving networks could provide further practical insights. By bridging strong edge coloring with Sierpiński graphs, this study contributes to combinatorial optimization and structured network design, opening new pathways for research in graph theory and applied computing.

References

- [1] O. Favaron, H. Li, and R. H. Schelp, “Strong Edge Colorings of Graphs,” *Discrete Mathematics*, vol. 159, no. 1, pp. 103–109, 1996. DOI: 10.1016/0012-365X(95)00102-3.
- [2] S. Akbari, H. Bidkhori, and N. Nosrati, “ r -Strong Edge Colorings of Graphs,” *Discrete Mathematics*, vol. 306, no. 23, pp. 3005–3010, 2006. DOI: 10.1016/j.disc.2004.12.027.
- [3] F. S. Benevides, C. Hoppen, and R. M. Sampaio, “Edge-Colorings of Graphs Avoiding Complete Graphs with a Prescribed Coloring,” *Discrete Mathematics*, vol. 340, no. 9, pp. 2143–2160, 2017. DOI: 10.1016/j.disc.2017.04.011.
- [4] L. Perković and B. Reed, “Edge Coloring Regular Graphs of High Degree,” *Discrete Mathematics*, vols. 165–166, pp. 567–578, 1997. DOI: 10.1016/S0012-365X(96)00202-6.
- [5] E. Andrews, L. Helenius, D. Johnston, J. VerWys, and P. Zhang, “On Twin Edge Colorings of Graphs,” *Discussiones Mathematicae Graph Theory*, vol. 3, 2014. DOI: 10.7151/dmgt.1756.
- [6] S. Dara, S. Mishra, N. Narayanan, et al., “Strong Edge Coloring of Cayley Graphs and Some Product Graphs,” *Graphs and Combinatorics*, vol. 38, no. 1, p. 51, 2022. DOI: 10.1007/s00373-021-02408-4.
- [7] H. Yin, M. Han, and M. Xu, “Strong Edge Coloring of $K(t)$ -Minor Free Graphs,” *Axioms*, vol. 12, no. 6, article 556, 2023. DOI: 10.3390/axioms12060556.
- [8] Z. Wang, D. Huang, and R. Pei, “Solving the Minimum Vertex Cover Problem with DNA Molecules in Adleman–Lipton Model,” *Journal of Computational and Theoretical Nanoscience*, vol. 11, pp. 521–523, 2014.
- [9] D. B. West, *Introduction to Graph Theory*, 2nd ed., Prentice Hall, 2001.
- [10] X. Yang, G. M. Megson, Y. Tang, and D. J. Evans, “Diameter of Parallelogramic Honeycomb Torus,” *Computers Mathematics with Applications*, vol. 50, pp. 1477–1486, 2005.

- [11] Y. Yigit, V. K. Akram, and O. Dagdeviren, "Breadth-first Search Tree Integrated Vertex Cover Algorithms for Link Monitoring and Routing in Wireless Sensor Networks," *Computer Networks*, vol. 194, article 108144, 2021.
- [12] Y. Zhang, S. Wang, C. Liu, and E. Zhu, "TIVC: An Efficient Local Search Algorithm for Minimum Vertex Cover in Large Graphs," *Sensors*, vol. 23, article 7831, 2023.
- [13] J. Anitha and I. Rajasingh, "Power Domination Parameters in Hypermesh Pyramid Networks and Corona Graphs," *International Journal of Pure and Applied Mathematics*, vol. 109, no. 5, pp. 59–66, 2016.
- [14] R. Barrera and D. Ferrero, "Power Domination in Cylinders, Tori, and the Generalized Petersen Graphs," *Networks*, pp. 43–49, 2009.
- [15] K. F. Benson, D. Ferrero, M. Flagg, V. Furst, L. Hogben, V. Vasilevska, and B. Wissman, "Zero Forcing and Power Domination for Graph Products," *arXiv preprint arXiv:1510.02421*, 2015. [Online]. Available: <https://arxiv.org/abs/1510.02421>.
- [16] B. Brimkov, D. Mikesell, and L. Smith, "Connected Power Domination in Graphs," *Journal of Combinatorial Optimization*, vol. 38, no. 1, pp. 292–315, 2019.
- [17] G. J. Chang, P. Dorbec, M. Montassier, and A. Rasput, "Generalized Power Domination of Graphs," *Discrete Applied Mathematics*, vol. 160, no. 12, pp. 1691–1698, 2012.
- [18] Z. Chen, S. Kosari, S. Omid, N. Mehdipoor, A. A. Talebi, and H. Rashmanlou, "Elementary Abelian Covers of the Wreath Graph $W(3,2)$ and the Foster Graph F_{26A} ," *AKCE International Journal of Graphs and Combinatorics*, vol. 20, pp. 20–28, 2022.
- [19] N. Alon and B. Sudakov, "Increasing the Chromatic Number of a Random Graph," *arXiv preprint arXiv:1002.1748*, 2010.
- [20] A. Akbulut and G. Yilmaz, "University Exam Scheduling System Using Graph Coloring Algorithm and RFID Technology," *International Journal of Innovation, Management, and Technology*, vol. 4, no. 1, 2013.
- [21] M. W. Carter and G. Laporte, "Recent Developments in Practical Examination Timetabling," in *International Conference on the Practice and Theory of Automated Timetabling*, 1995, pp. 1–21.
- [22] R. Ganguli and S. Roy, "A Study on Course Timetable Scheduling Using Graph Coloring Approach," *International Journal of Computational and Applied Mathematics*, vol. 12, no. 2, pp. 469–485, 2017.
- [23] J. Matoušek and G. M. Ziegler, "Topological Lower Bounds for the Chromatic Number: A Hierarchy," *arXiv preprint math/0208072*, 2002.
- [24] N. Robertson, D. Sanders, P. Seymour, and R. Thomas, "The Four-Colour Theorem," *Journal of Combinatorial Theory, Series B*, vol. 70, pp. 2–44, 1997.
- [25] A. Ilić and M. Bašić, "On the Chromatic Number of Integral Circulant Graphs," *Computers & Mathematics with Applications*, vol. 60, no. 1, pp. 144–150, 2010.